TEAM INFERENCE SQUAD

PREDICTING NCAA TOURNAMENT SEEDS

A Machine Learning Approach to Quantifying Committee Judgment

Indiana University Bloomington | NCAA Final Four Analytics Challenge Finals

April 6, 2026

The Divergence Problem

When NET rank and seed diverge sharply, it reveals unmeasured committee judgment. The gap our model was built to close.

Colgate 2024-25 · Patriot League

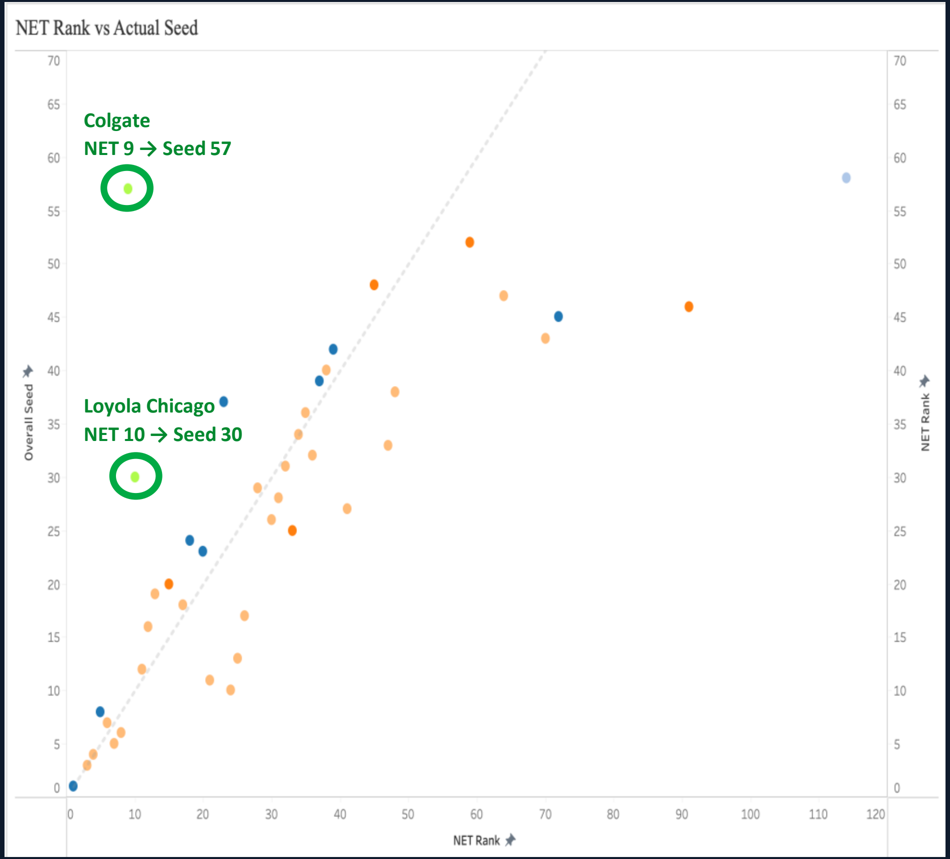
NET Rank #9 → Seed #57 (48 positions off)

Loyola Chicago 2020-21 · MVC

NET Rank #10 → Seed #30 (20 positions off)

Top-10 NET efficiency. Seeded 57th.
Same metrics. Same system. The committee is applying judgment the data alone can't explain.

Conference affiliation is the hidden variable.



The Committee's Challenge

362 Division I programs. 31 conferences.

Not equal — and not static.

Power Conferences

12+ bids in a single season.

Opponents ranked inside the top 30 every night.

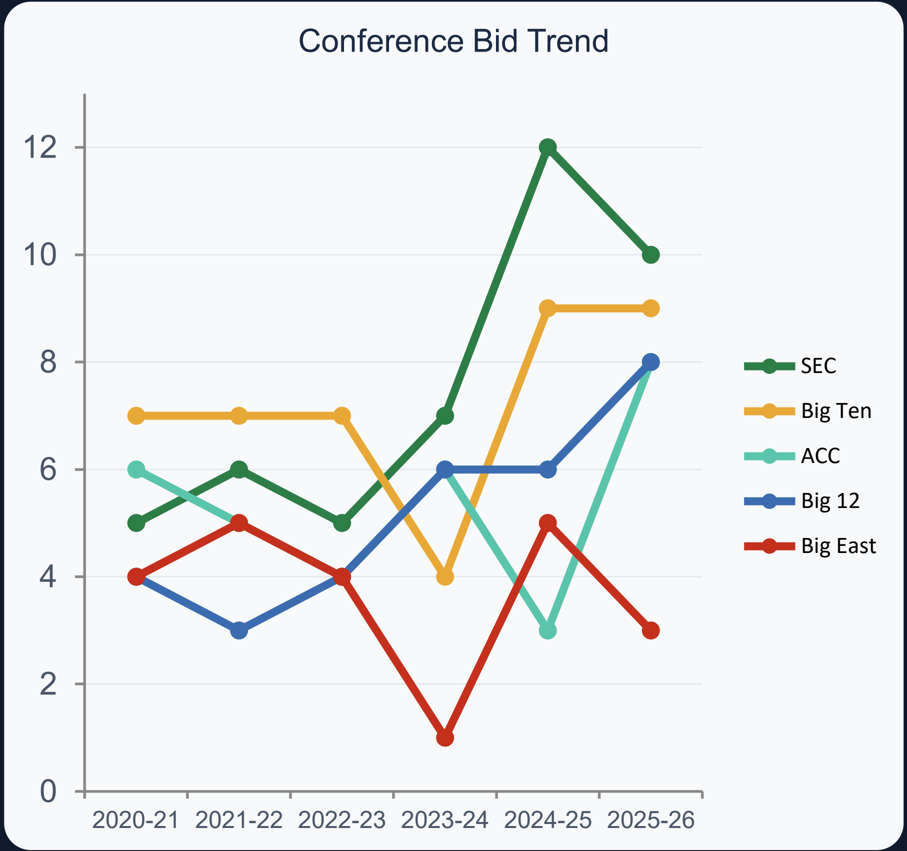
Mid / Low-Major

One or two bids — if that.

One guaranteed spot: conference champion only.

The SEC surged to 12 bids in 2024-25.
The ACC fell from 6 to 3.
The committee must recalibrate every year.

How do you fairly compare 10-3 in the SEC vs 16-1 in the Patriot League?



THE OBJECTIVE

Predict tournament seeds 1–68 using the same metrics the committee already has.

1	**Consistency Check**	Flag the 15–20 cases that diverge most from 5-year historical patterns
2	**Public Confidence**	Explicit rationale when the committee departs from the model
3	**Pattern Discovery**	Are mid-major AQ teams systematically under-seeded?
4	**Auto-Improving**	Retrains each season — a living record of committee behavior
5	**Cognitive Load ↓**	Focus deliberation time on the true bubble

*We are not here to replace the committee — **we are here to give them a sharper, data-driven lens.***

DNA of a Champion

What five years of radar data reveals about the teams that make it deep

Q3 / Q4 Win %

Top seeds are near-perfect against weaker opponents, Q3 and Q4 win rates close to 100%.

Lower seeds and non-tournament teams drop off sharply. Bad losses reveal who doesn't belong.

Bad Losses

Near zero for 1-seeds. Sharply higher as you move down the bracket.

Championship teams don't lose to the worst.

SOS + Road Wins

Strength of schedule and road win % separate bubble teams from locks.

Winning away — against elites — is what the committee weights most.

"Championship-caliber teams win against the best. And they never lose to the worst."

The Committee Logic Advantage

We built 14 features from scratch by studying 5 years of committee behavior.

13.5% of features → 28% of predictive power

Committee Logic features punch nearly double their proportional share.

Top engineered features:

#2 AQ_Conf_Penalty

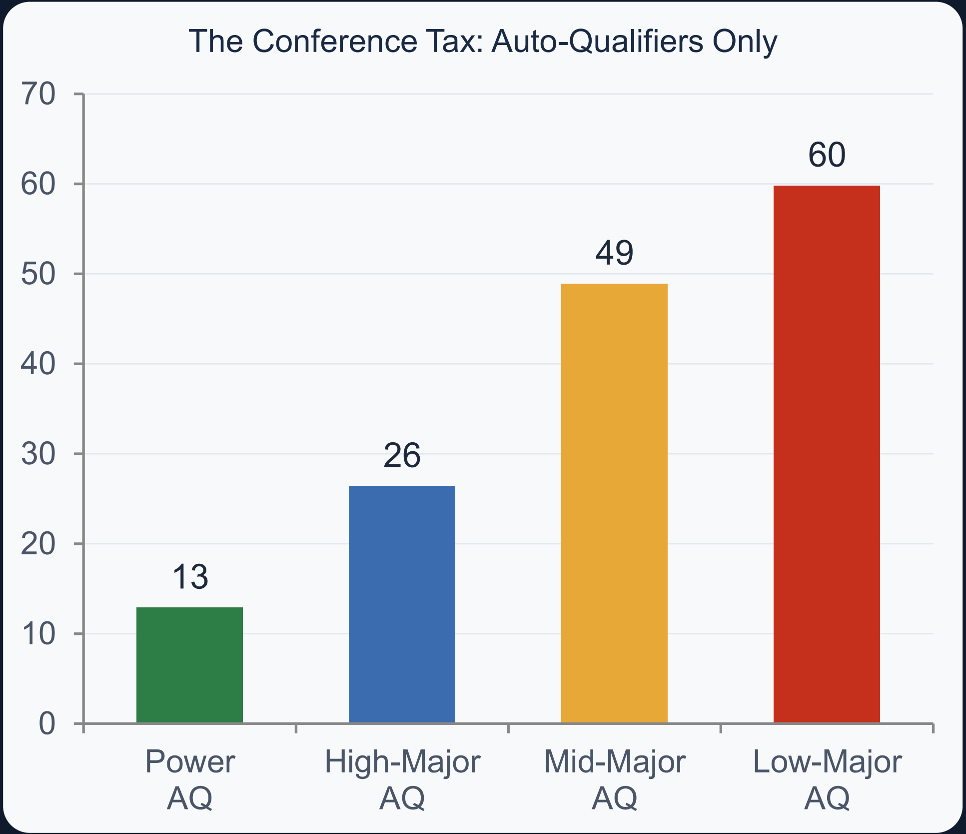
#4 AL_Conf_Strength

#6 Conf_Avg_NET

None of these exist in any NCAA database. We built them.

RMSE: **2.39** → **1.69** → **1.37**

Biggest drop came from encoding human judgment



What the Model Delivers

Out-of-sample results on 2025-26 data the model had never seen

96.26%	77.9%	3.48	91.8%
R-Squared	**Selection Recall**	**Matched RMSE**	**Overall Accuracy**
Model explains 96% of variance in seeding	53 of 68 tournament teams correctly identified	Avg seed error for correctly identified teams	335 of 365 teams correctly classified

75.5% of correctly identified teams within 4 seed positions of actual

Baseline (NET Rank alone): RMSE 19.12 Our model: RMSE 3.48 Improvement: 82%

8	21	11	12	16
Exact	Within 2	Within 4	Within 8	9+ off

MODEL ARCHITECTURE · 7-MODEL ENSEMBLE

HistGradientBoosting ×3
varying learning rates + subsampling strategies

GradientBoostingRegressor ×1

ExtraTreesRegressor ×2

Tournament-Only Specialist ×1
captures fine-grained seed ordering

→ **Ensemble average output**

→ Constrained post-processing: exactly 68 unique seeds

→ Uniqueness constraint via iterative rank adjustment

→ Tournament-only models blended for seed precision

NET Rank RMSE baseline: 19.12 → **Model RMSE: 3.48** (−82% reduction)

104 features | 3 stages (22 → 60 → 72 → 102) | 7 models | < 60 sec runtime

THE PIPELINE — END TO END

RAW DATA
Season stats CSV
20 cols / team

→

CLEAN
Fix Excel date
corruption

→

FEATURES
104 features
across 3 stages

→

ENSEMBLE
7-model
blend

→

CONSTRAIN
Exactly 68
unique seeds

→

PREDICT
Full bracket
< 60 seconds

New season = new CSV. No manual tuning. No hand-picking.

Model improves automatically as each tournament adds to the historical record.

2025-26 OUT-OF-SAMPLE RESULTS

Team	Predicted	Actual	Δ / Note
Duke	1	1	✓ Exact
Florida	4	4	✓ Exact
Auburn	2	3	±1
Houston	5	6	±1
Gonzaga	6	11	±5 — WCC strength underweighted
Louisville	16	23	±7 — resume gap underestimated

53 / 68
Teams Correctly Selected

77.9%
Selection Recall

75.5%
Within 4 Seed Positions

WHAT COMES NEXT

Cloud Pipeline

End-to-end automated retraining. Each year's unseen data becomes training data, so the model learns evolving committee decisions automatically

Game-Level Data

Margin of victory · recency weighting · late-season momentum signals

Real-Time NET

Live seed projections as conference tournaments unfold, not end-of-season snapshots only

A tool the committee can use before Selection Sunday, not to override judgment, but to sharpen it.

Thank you